

- 7 (a) Fig. 7.1 shows a red deer, *Cervus elaphus*. Red deer are herbivores that feed on a wide range of plants.



Fig. 7.1

The number of red deer in the UK increased between 1960 and 2010, as shown in Table 7.1.

Table 7.1

year	number
1960	135 000
1970	180 000
1980	250 000
1990	300 000
2000	330 000
2010	360 000

- (i) Calculate the percentage increase in red deer in the UK from 1960 to 2010.

Give your answer to one decimal place.

answer %
[2]

- (ii) Environmental factors affect the population size of red deer so that numbers do not continue to increase forever.

Suggest environmental factors that may prevent further increase in the size of a red deer population.

.....

.....

.....

.....

.....

.....

.....[3]

- (b) The body mass of red deer shows wide variation within populations and this variation is shown in Fig. 7.2.

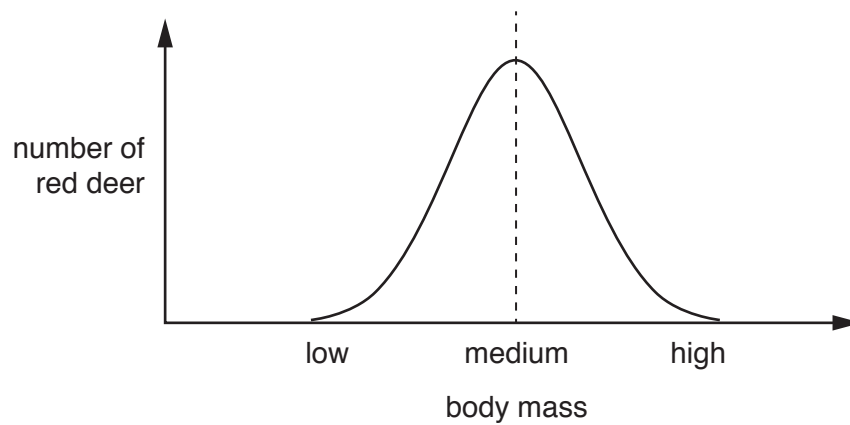


Fig. 7.2

- (i) A selection pressure acted consistently over many years against red deer of **low** body mass in a population.

Sketch a curve on Fig. 7.3 to show the pattern of variation of body mass in this red deer population after this time **and** name the type of force of natural selection that is acting.

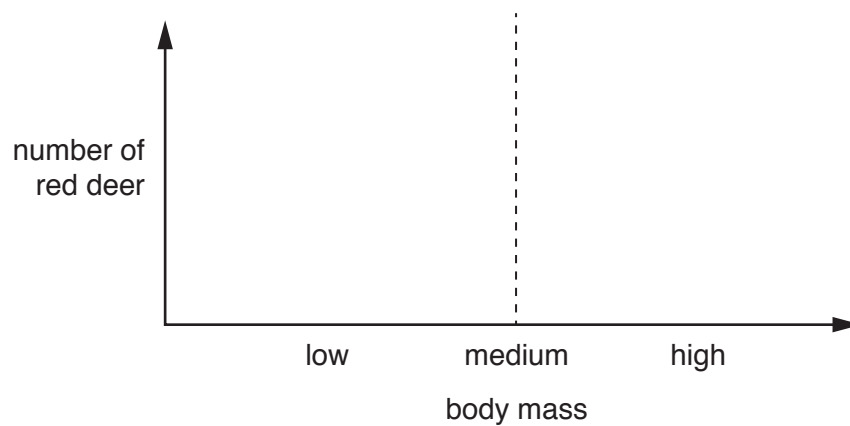


Fig. 7.3

type of force of natural selection [2]

- (ii) A selection pressure acted consistently over many years against red deer of **medium** body mass in a different population.

Sketch a curve on Fig. 7.4 to show the pattern of variation of body mass in this red deer population after this time **and** name the type of force of natural selection that is acting.

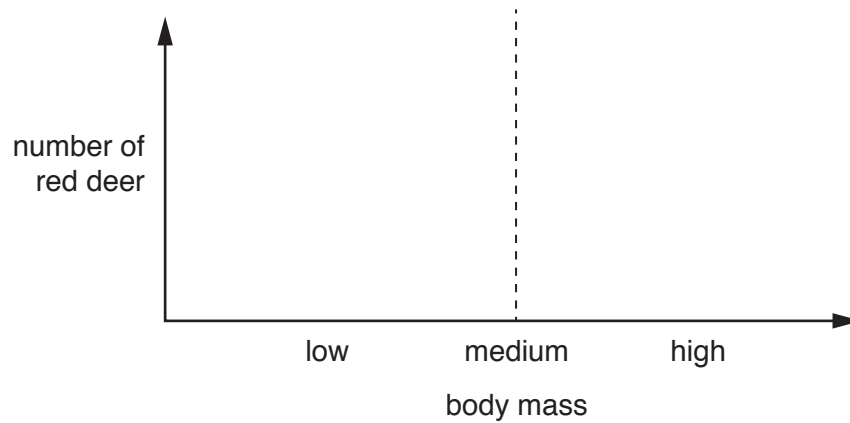


Fig. 7.4

type of force of natural selection [2]

[Total: 9]